GREPHOUND / MEASURED ECONOMICS

A cheaper scout cut the measured bill.

GPT-5.6 Luna searched the repository before GPT-5.6 Sol answered. The useful result is not “fewer tokens” by itself. It is less Sol work and a lower complete provider cost after adding Luna.

39.2%	$0.388	$0.265	$0.0098
median complete-cost reduction	median direct cost	median assisted cost	median Luna scout cost

What the median run actually cost

	Sol alone	Grephound	Change
Complete provider bill	$0.387586	$0.265499	**-$0.122087**
Sol portion	$0.387586	$0.255668	**-$0.131918**
Luna scout portion	—	$0.009831	+$0.009831
Sol input tokens	98,684	77,878	**-21.08%**
Luna input tokens	—	78,830	+78,830
All-model input tokens	98,684	156,708	**+58.80%**
Quality checks	6/6	6/6	tied

The mechanism: Grephound added about one cent of Luna work, then removed about thirteen cents of Sol work. Total input rose in this median pair because Luna did the search. The bill still fell because Luna tokens were much cheaper.

Three repeated pairs, one repository question

Pair	Direct	Assisted	Luna	Dollars saved	Cost change
1	$0.372301	$0.226347	$0.009905	**$0.145954**	**-39.20%**
2	$0.387586	$0.265499	$0.009831	**$0.122087**	**-31.50%**
3	$0.832314	$0.304628	$0.009548	**$0.527686**	**-63.40%**

All three pairs passed the same 6/6 behavior rubric in both arms. Median paired cost change was -39.20%. Median wall time increased 31.21%; pair 3 was faster assisted. These are repeated diagnostics on one read-only question, not a universal product savings claim.

TOKEN AND COST ACCOUNTING

Luna tokens are visible, not hidden.

Every assisted total below includes both models. Luna pricing used \$0.20/M uncached input, \$0.02/M cached input reads, and \$1.20/M output including reasoning. Cache writes were unbilled.

The repeated optimized cohort

	Sol alone		Grephound assisted			
Pair	Sol input	Cost	Sol input	Sol cost	Luna input	Luna cost
1	110,033	\$0.372301	80,522	\$0.216442	77,310	\$0.009905
2	98,684	\$0.387586	77,878	\$0.255668	78,830	\$0.009831
3	437,052	\$0.832314	66,630	\$0.295080	76,660	\$0.009548

Luna input includes cached and uncached input as reported. Cost is cache-aware; raw token count is not used as a cost proxy. Each assisted run used one scout conversation, three repository tool calls, and five validated citations.

The real coding-task diagnostic

\$0.8694
Sol alone

\$0.4336
Sol + Luna

\$0.4358
provider dollars saved

50.12%
complete-cost reduction

Astropy SWE-bench Verified	Sol alone	Sol + Luna	Change
Sol input tokens	823,105	324,636	**-60.56%**
Luna input tokens	—	62,766	+62,766
Luna cached input	—	40,544	+40,544
Luna output tokens	—	1,642	+1,642
Luna cost	—	\$0.008336	+\$0.008336
Sol cost	\$0.869422	\$0.425304	**-\$0.444118**
Complete provider bill	\$0.869422	\$0.433640	**-\$0.435782**
All-model tokens	831,166	394,780	**-52.50%**
Wall time	426.52 s	385.57 s	**-9.60%**
Verification	pass	pass	tied

Task: fix Astropy's ASCII HTML writer so write(formats=...) formats HTML table output. Both arms passed the exact gold regression and the full HTML test file. The Luna scout returned the writer, formatting path, tests, and focused file/line evidence. This is one stochastic task using the preserved FastContext explorer, not the Grephound Rust runtime and not headline evidence.

DECISION AND LIMITS

What the evidence supports now.

Supported

Claim	Evidence
Complete cost fell	3/3 repeated quality-passing pairs cost less assisted; median paired reduction 39.20%.
Sol work fell	Median Sol input fell 26.82% in the optimized cohort and 50.19% under immediate routing.
Luna was cheap enough	Median Luna scout cost was $0.009831 in the optimized cohort and $0.008478 under immediate routing.
The handoff was useful	Each optimized scout used three repository tools and returned five validated citations; the main agent did no repository search after handoff.

Not supported yet

No universal savings percentage. The paired economics repeat one repository question.
No claim that total tokens always fall. In the optimized median pair, all-model input rose 58.80% while cost fell.
No latency win. Median wall time rose 31.21% in the optimized cohort and 6.65% under immediate routing.
No local-model win. FastContext 4B Q4 produced only one usable handoff in three bounded runs and remains disabled by default.

Current production choice

Setting	Retained value
Routing	Scout first only for eligible, unknown-location broad exploration
Scout	GPT-5.6 Luna, medium reasoning
Bound	120 seconds; three repository tools
Handoff	Five citations; 6 KiB evidence cap
Repository tool result cap	32 KiB
Native repository map	Removed after median Luna cost increased 20.53%
Local FastContext default	Disabled as unreliable

Next evidence gate

34 independent tasks × 3 randomized, blind-graded repeats per arm.

Only after that run should the project publish a general cost-reduction headline. The gate requires complete Sol + Luna accounting, quality grading before economics, preserved losing runs, and provider-bill totals rather than a middleware savings counter.